NET731 Advanced Networking

Week 6 Practical: Cloud Security Review

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Module: NET731

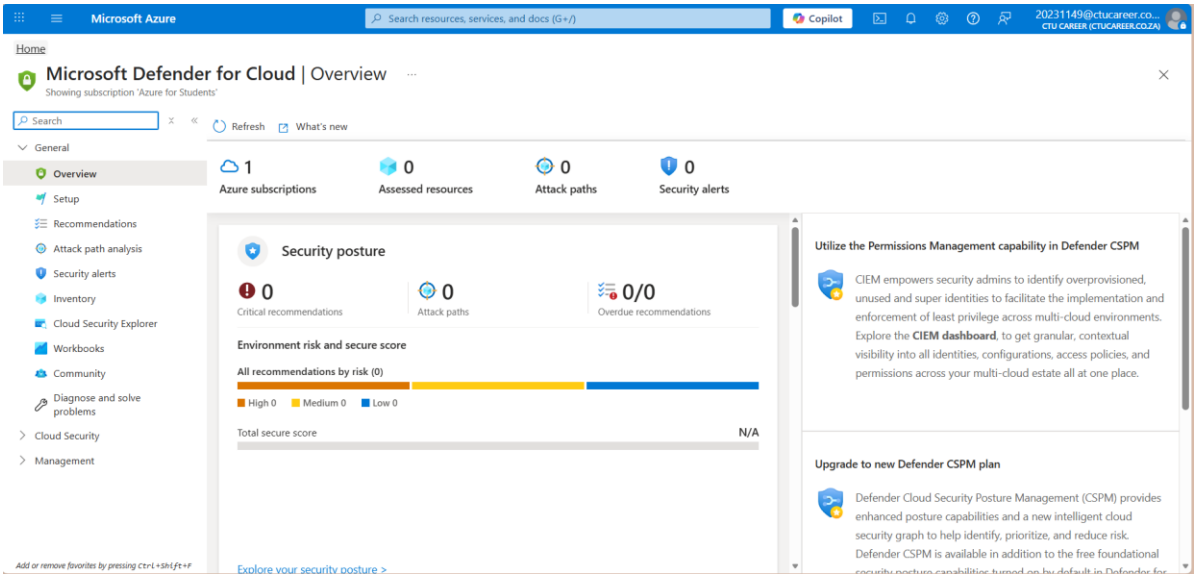
1. Introduction

The purpose of this practical was to review the security posture of the Azure environment, identify risks, and apply targeted remediations. Microsoft Defender for Cloud was used as the starting point for the security review, and the remediation work focused on two required areas: closing open management ports and enabling operating system update assessment on the deployed virtual machine.

2. Security Posture Assessment

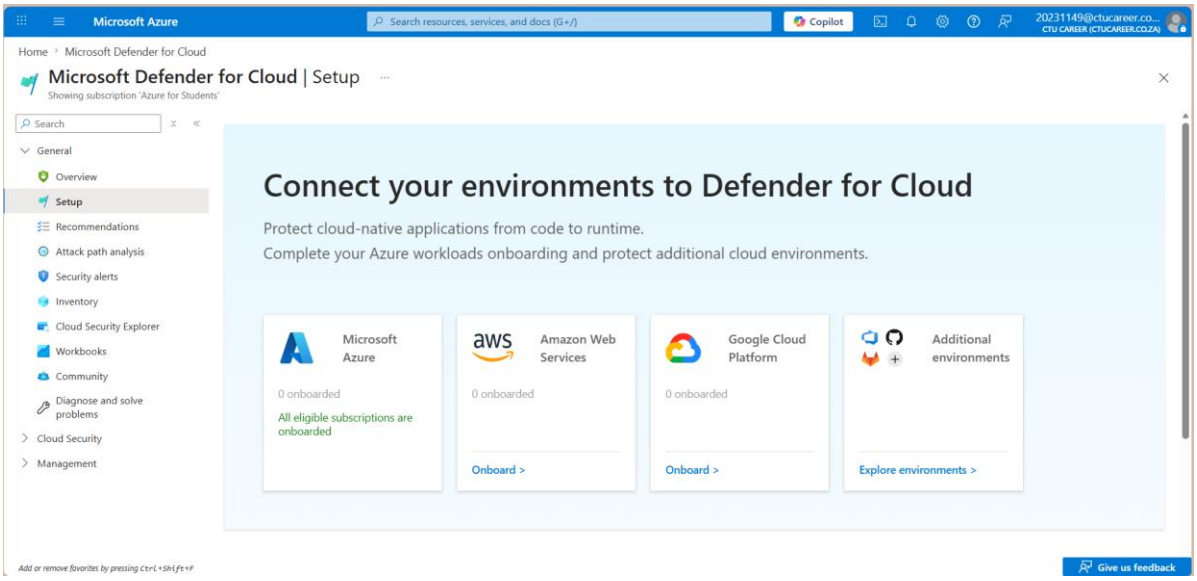
2.1 Microsoft Defender for Cloud Overview

Microsoft Defender for Cloud was opened from the subscription. The dashboard showed one Azure subscription, zero assessed resources, zero active attack paths, zero security alerts, and a Secure Score displayed as N/A. This indicated that the environment was onboarded, but Defender for Cloud was not actively surfacing assessed resources or recommendation data for the deployed environment at that point.



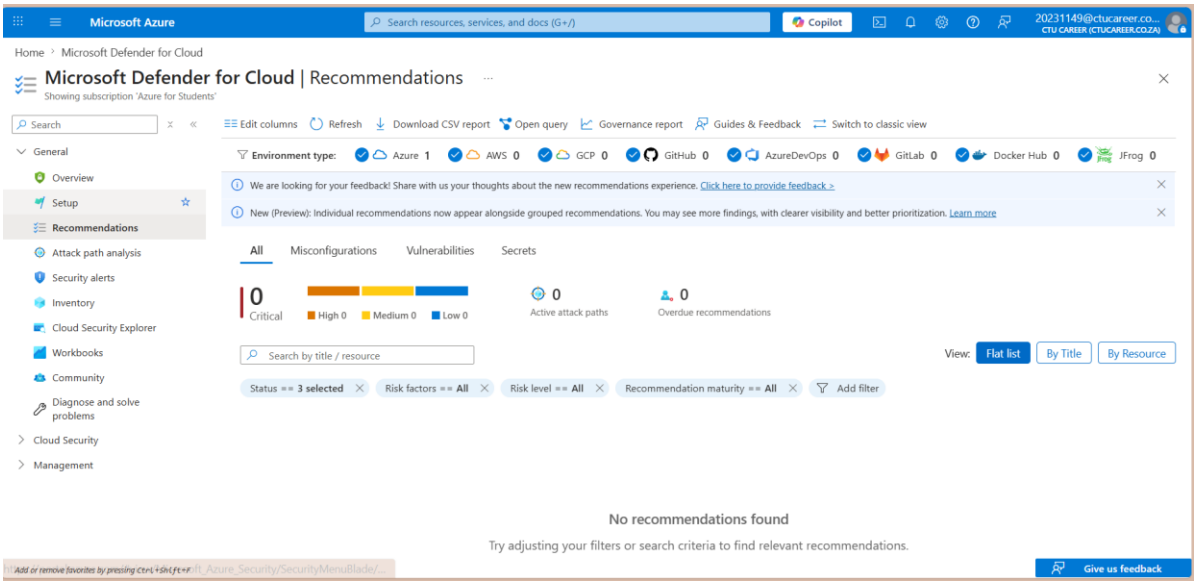
2.2 Defender Setup Status

The Defender for Cloud setup page showed that the Azure subscription was there and that all eligible subscriptions were shown. Even though the environment appeared to be connected, the service still reported zero assessed resources.



2.3 Recommendations Review

The recommendations page was reviewed to identify the highest-risk findings. However, the recommendations list returned no active findings, zero critical, zero high, zero medium and zero low recommendations. Because no active recommendations were shown, there were no top three recommendations to analyse directly.



3. Key Security Findings

The security assessment identified three important findings about the current environment:

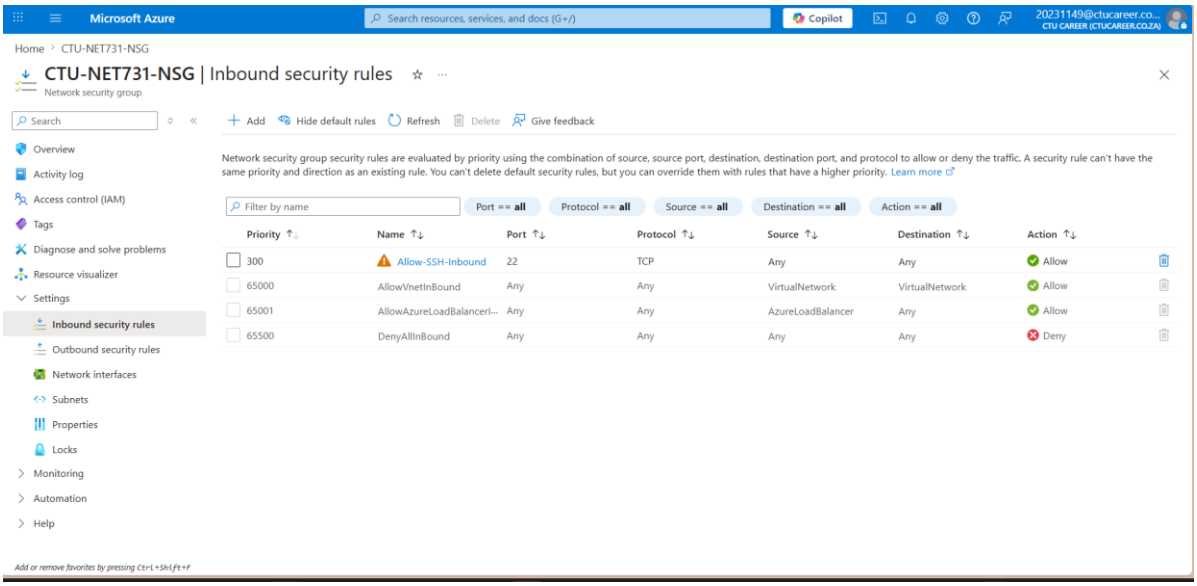
- Microsoft Defender for Cloud was available, but the Secure Score was shown as N/A.
- The environment reported zero assessed resources, which meant Defender for Cloud did not provide actionable recommendation data.
- The Azure environment still contained a custom inbound NSG rule that allowed SSH (port 22) from any source, which represented a real management port exposure and could be remediated directly.

4. Close Open Management Ports

The first required remediation was to remove any open management port exposed through the Network Security Group. The inbound rules for CTU-NET731-NSG were reviewed and the custom rule Allow-SSH-Inbound was found. This rule allowed TCP port 22 traffic from any source to any destination. Because open SSH access is a common attack vector, the rule was deleted to reduce the attack surface of the virtual machine.

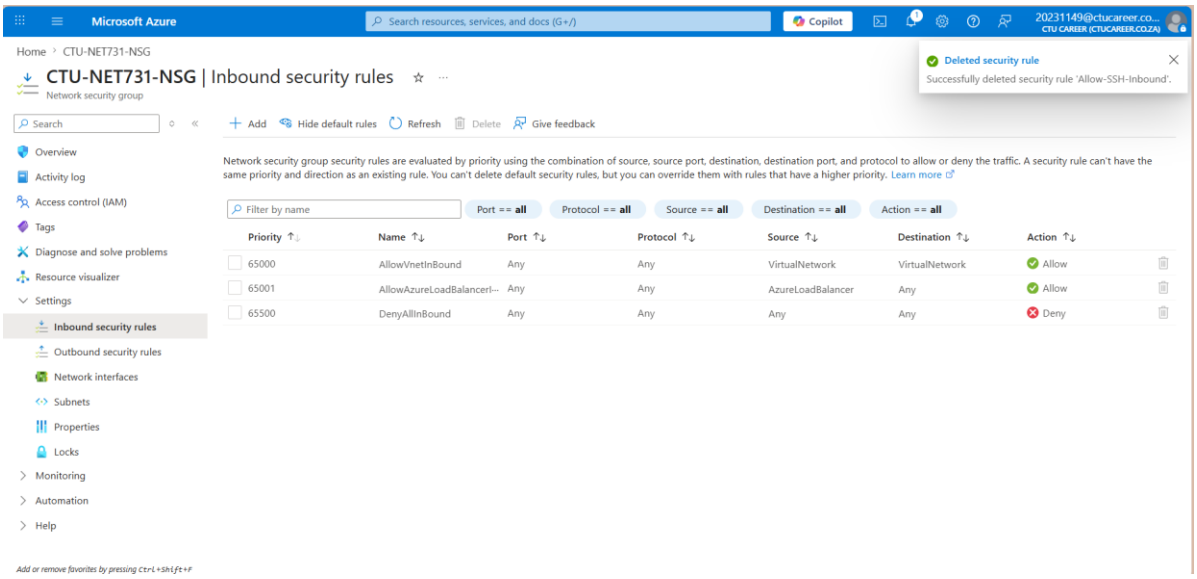
4.1 Before remediation

Before the change, the NSG contained the custom rule Allow-SSH-Inbound on priority 300 with port 22 and action Allow.



4.2 After remediation

After the change, the custom SSH rule was removed successfully, leaving only the default inbound security rules. This closed public SSH access to the virtual machine.

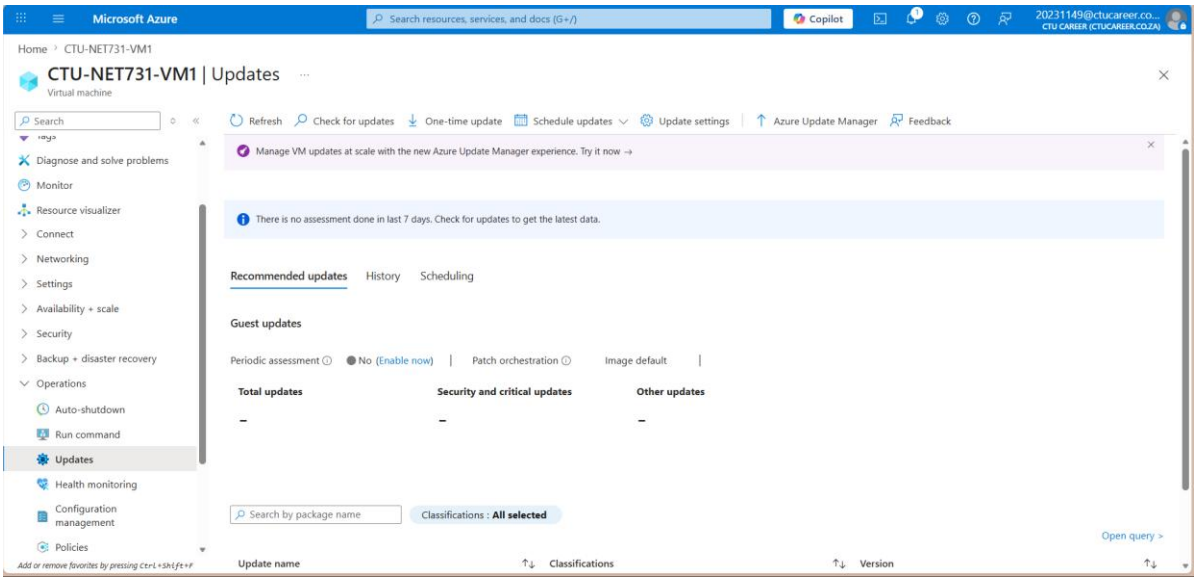


5. Enable OS Updates

The second required remediation was to enable operating system update assessment for the deployed virtual machine CTU-NET731-VM1. The Updates page for the VM was reviewed first to confirm the initial state. The page showed that periodic assessment was set to No, meaning regular guest update assessment was not enabled. Azure Update Manager settings were then opened, and the Periodic assessment option was changed from Disable to Enable. After saving the configuration, the VM Updates page showed Periodic assessment = Yes, confirming that the setting had been applied.

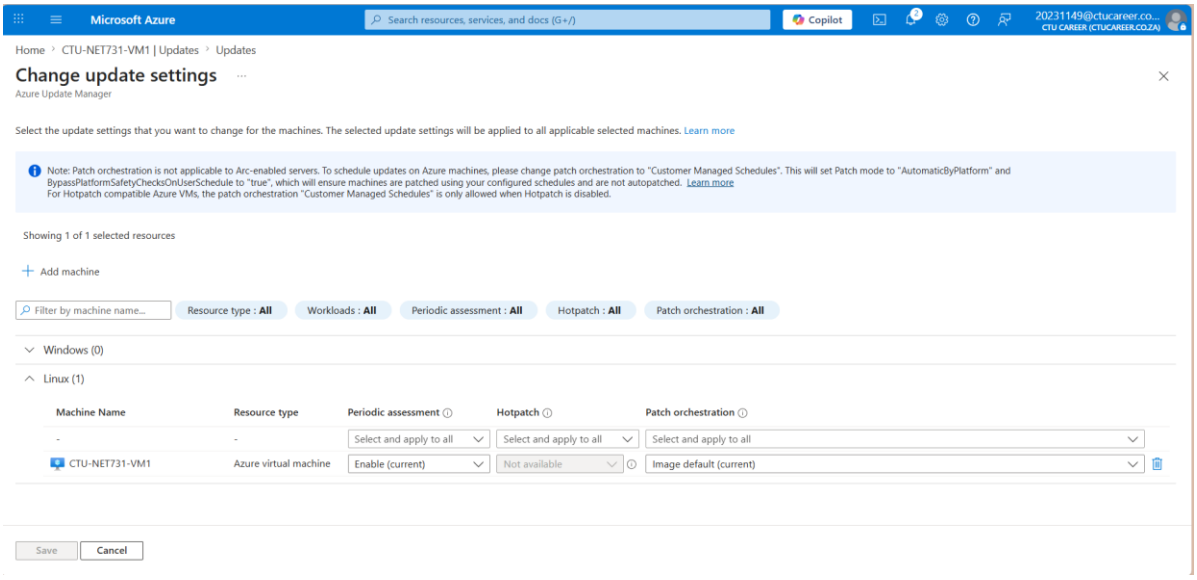
5.1 Before remediation

The updates page showed that periodic assessment was disabled.



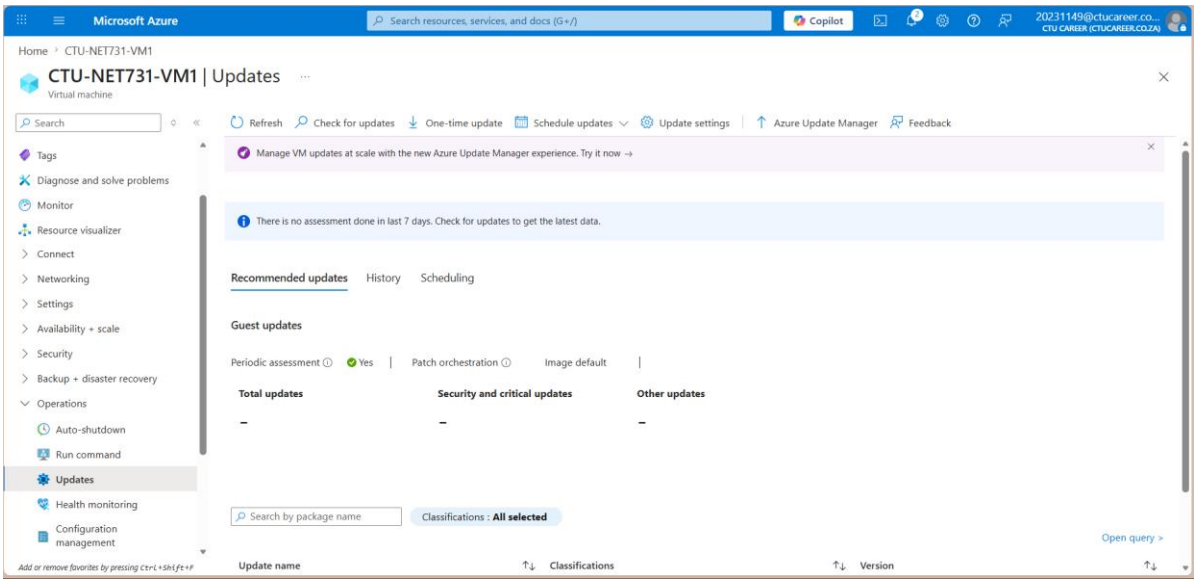
5.2 Configuration applied

On the Azure Update Manager change page, the periodic assessment option was enabled.



5.3 After remediation

After saving the setting, the Updates page confirmed that periodic assessment was now enabled for the VM.



6. Security Impact of the Changes

The security improvements applied in this practical reduced two important risks in the Azure environment. Removing the custom SSH allow rule reduced unnecessary exposure of the virtual machine to the public internet and lowered the risk of unauthorised remote access. Enabling periodic update assessment improved visibility into missing system updates and supports better patch management for the virtual machine. Together, these changes improved the overall security posture of the environment even though Defender for Cloud was not actively surfacing recommendations for the student subscription.

7. LMS Submission Summary

The security finding in my environment was the open SSH management port that had been allowed through the Network Security Group. I prioritised this issue first because management ports exposed to the public internet create a direct path for unauthorised access attempts against the virtual machine. After the review, I removed the Allow SSH Inbound rule and then enabled periodic update assessment on the VM so that missing updates can be monitored more effectively. These two changes addressed the most important practical security risks that were visible in the Azure student environment.